torch.signal.windows.cosine#

https://pytorch.org/docs/stable/generated/torch.signal.windows.cosine.html

Description:

Computes a window with a simple cosine waveform, following the same implementation as SciPy. This window is also known as the sine window. The cosine window is defined as follows: This formula differs from the typical cosine window formula by incorporating a 0.5 term in the numerator, which shifts the sample positions. This adjustment results in a window that starts and ends with non-zero values. The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and sym is True. M (int) – the length of the window. In other words, the number of points of the returned window.

Function Signature

```
torch.signal.windows.cosine(M, *, sym=True, dtype=None,
layout=torch.strided, device=None, requires_grad=False)
[source]#

Parameters

Keyword Arguments

Return type
```

Parameters

Keyword

sym (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True. dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()). layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided. device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types. requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Defaul

Returns:

Tensor

Code Examples

```
>>> # Generates a symmetric cosine window.
>>> torch.signal.windows.cosine(10)
tensor([0.1564, 0.4540, 0.7071, 0.8910, 0.9877, 0.9877, 0.8910,
0.7071, 0.4540, 0.1564])

>>> # Generates a periodic cosine window.
>>> torch.signal.windows.cosine(10, sym=False)
tensor([0.1423, 0.4154, 0.6549, 0.8413, 0.9595, 1.0000, 0.9595,
0.8413, 0.6549, 0.4154])
```

Detailed Documentation

Documentation

Computes a window with a simple cosine waveform, following the same implementation as SciPy. This window is also known as the sine window.

The cosine window is defined as follows:

This formula differs from the typical cosine window formula by incorporating a 0.5 term in the numerator, which shifts the sample positions. This adjustment results in a window that starts and ends with non-zero values.

The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and sym is True.

M (int) – the length of the window. In other words, the number of points of the returned window.

sym (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True.

dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()).

layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided.

device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.